

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AC1-163***

***Ahoskie 34.5kV
12.6 MW Capacity / 16.9 MW Energy***

December / 2017

Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, Section 205, as well as the System Impact Study Agreement between sPower Development Company, LLC, the Interconnection Customer (IC) and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Virginia Electric and Power Company (VEPCO).

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the IC. As a requirement for interconnection, the IC may be responsible for the cost of constructing Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an IC may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The IC is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

The IC has proposed a solar generating facility located in Hertford County, NC. The installed facilities will have a total capability of 16.9 MW with 11.5 MW of this output being recognized by PJM as capacity. The proposed in-service date for this project is 10/01/2019. **This study does not imply an ITO commitment to this in-service date.**

Attachment Facilities and local upgrades (if required) along with terms and conditions to interconnect AC1-163 will be specified in a separate two party Interconnection Agreement (IA) between ITO and the IC as this project is considered FERC non-jurisdictional per the PJM Open Access Transmission Tariff (OATT). From the transmission system perspective, no network impacts were identified as detailed below.

Point of Interconnection

AC1-163 will interconnect with the ITO distribution system via a tap on to the Ahoskie 34.5kV circuit.

Attachment facilities and local upgrades (if required) along with terms and conditions to interconnect AC1-163 will be specified in a separate two party Interconnection Agreement (IA) between ITO and the IC as this project is considered FERC non-jurisdictional per the PJM Open Access Transmission Tariff (OATT). This work includes ITO's System Protection Engineering Department to review transmission line protection as well as anti-islanding required to accommodate the new generation and interconnection substation. System Protection Engineering will determine the minimal acceptable protection requirements to reliably interconnect the proposed generating facility with the transmission system.

Cost Summary

The AC1-163 interconnection request will be responsible for the following costs:

Description	Total Cost
Transmission Owner facilities	\$0
Allocation for Transmission Upgrades	\$54,854
Total Costs	\$54,854

Transmission Owner Scope of Work

An express 34.5kV line will need to be built to the IC's site and the requested POI because of the close proximity to the substation. The proposed 34.5 kV express feed, will be served from a new 56 MVA, 115/34.5 kV transformer in Ahoskie Substation.

Non-Direct Connection Cost Estimate

New System Reinforcements

PJM OATT 217.3 outlines cost responsibility for Network Upgrades and as the minimum amount of Network Upgrades required to resolve a single reliability criteria violation will not meet or exceed \$5,000,000 such costs shall be allocated to those Interconnection Requests in the New Services Queue that contribute to the need for such upgrades. Such allocations shall be made in proportion to each Interconnection Request's megawatt contribution to the need for these upgrades subject to the rules for minimum cost allocation thresholds in the PJM Manuals. For the purpose of applying the \$5,000,000 threshold, each reliability criteria violation shall be considered separately.

Reinforcement: Line #254 AB2-100 Tap – Clubhouse 230 kV overload: Rebuild the AB2-100 TAP-Clubhouse 230kV line to increase its line rating to 706 MVA (normal), 706 MVA (emergency), and 812 MVA (load dump). It is estimated to take 24-28 months to permit and construct a Virginia CPCN will most likely be required for this rebuild, the estimated cost is \$5,000,000 to resolve this deficiency.

Queue	MW contribution	Percentage of Cost	Cost (\$5M)	Contingency Name	Contingency Type
AB2-169	7.59	3.32%	\$165,872.63	'246T2034_A'	breaker
AC1-034	14.65	6.40%	\$320,162.59	'246T2034_A'	breaker
AC1-054	7.02	3.07%	\$153,415.80	'246T2034_A'	breaker
AC1-086	179.79	78.58%	\$3,929,149.00	'239T2141'	breaker
AC1-099	17.23	7.53%	\$376,546.18	'246T2034_A'	breaker
AC1-163	2.51	1.10%	\$54,853.80	'LN 246_A'	single

Reinforcement: Battleboro – Rocky Mt 115kV: Replace Battleboro substation terminal equipment. Upgrading the breaker leads at Battleboro will bring the rating to 239/239/239 MVA. Estimated cost is \$15,000.

Note: Duke/Progress Energy portion of this line will need to be studied under Duke's FERC tariff process.

IC needs to enter into an Affected System Facilities Study agreement with Duke / Progress Energy (DEP) to determine how to mitigate Battleboro - Rocky Mt.115kV overload. The upgrade will likely be a complete reconductor, probably replacing structures. Reconductoring the circuit to 1590 ACSR will bring the rating to 185/205/214 MVA (Limited by Breaker Leads at Battleboro). Upgrading the terminal equipment at Rocky Mount will bring the rating to 311/311/311 MVA (Limited by 1590 ACSR conductor). The estimated cost \$15 million with an ISD of 6/1/2020.

Interconnection Customer Requirements

ITO's Facility Interconnection Requirements as posted on PJM's website

<http://www.pjm.com/~media/planning/plan-standards/private-dominion/facility-connection-requirements1.ashx>

Meteorological Data Reporting Requirement - The solar generation facility shall, at a minimum, be required to provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

Interconnected Transmission Owner Requirements

Metering and SCADA/Communication equipment must meet the requirements outlined in section 3.1.6 Metering and Telecommunications of ITO's Facility Connection Requirement NERC Standard FAC-001 which is publically available at www.dom.com.

Network Impacts

The Queue Project AC1-163 was evaluated as a 16.9 MW (Capacity 12.6 MW) injection at the Ahoskie 115kV substation in the ITO area. Project AC1-163 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AC1-163 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Contingency Descriptions

The following contingencies resulted in overloads:

Contingency Name	Description
3CAROLNA-6CAROLNA	CONTINGENCY '3CAROLNA-6CAROLNA' OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 END
LN 2058-2181	CONTINGENCY 'LN 2058-2181' OPEN BUS 304226 /* ISLAND: 6PA-RMOUNT#4115.00 OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1 /* 6PA- RMOUNT#4230.00 - 6NASH 230.00 OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1 /* 6HATHAWAY 230.00 - 6NASH 230.00 OPEN BUS 314591 /* ISLAND: 6NASH 230.00 OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1 /* 6ROCKYMT230T230.00 - 6HATHAWAY 230.00 END
LN 246_A	CONTINGENCY 'LN 246_A' OPEN BRANCH FROM BUS 921571 TO BUS 314575 CKT 1 /* AA1-138 TAP - 6NUCO TP 230.00 OPEN BRANCH FROM BUS 314569 TO BUS 314575 CKT 1 /* 6EARLEYS 230.00 - 6NUCO TP 230.00 OPEN BRANCH FROM BUS 314575 TO BUS 314590 CKT 1 /* 6NUCO TP 230.00 - 6NUCOR 230.00 OPEN BUS 314575 /* ISLAND OPEN BUS 314590 /* ISLAND END
LN 246_B	CONTINGENCY 'LN 246_B' OPEN BRANCH FROM BUS 314537 TO BUS 921571 CKT 1 /* 6SUFFOLK 230.00 - AA1-138 TAP END

Summer Peak Analysis – 2020

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

#	Contingency		Affected Area	Facility Description	Bus			Power Flow	Loading %		Rating		MW Contribution	Ref
	Type	Name			From	To	Cir.		Initial	Final	Type	MVA		
1	N-1	LN 246_A	DVP - DVP	AB2-100 TAP-6CLUBHSE 230 kV line	924510	314563	1	AC	99.91	100.57	ER	375	2.51	1

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output).

None

Short Circuit

(Summary of impacted circuit breakers)

New circuit breakers found to be over-duty:

None

Contributions to previously identified circuit breakers found to be over-duty:

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

#	Contingency	Affected	Facility Description	Bus	Cir.	Power	Loading %	Rating	MW	Ref
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	Type	Name	Area		From	To		Flow	Initial	Final	Type	MVA	Contribution	
2	DCTL	LN 2058-2181	DVP - CPLE	3BTLEBRO-3ROCKYMT115T 115 kV line	314554	304223	1	AC	156.84	157.85	ER	164	1.7	2

Steady-State Voltage Requirements

(Summary of the VAR requirements based upon the results of the steady-state voltage studies)

None

Stability and Reactive Power Requirement for Low Voltage Ride Through

(Summary of the VAR requirements based upon the results of the dynamic studies)

Not required

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this interconnection request)

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AC1-163 Allocation
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Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AC1-163 Allocation																					
# 1	AB2-100 TAP-6CLUBHSE 230 kV line	Rebuild the AB2-100 TAP-Clubhouse 230kV line #254 to increase its line rating to 706 MVA (normal), 706 MVA (emergency), and 812 MVA (load dump). It is estimated to take 24-28 months to permit and construct a Virginia CPCN will most likely be required for this rebuild. <table><tr><th>Queue</th><th>Impact (MW)</th><th>Cost</th></tr><tr><td>AB2-169</td><td>7.59</td><td>\$181,536</td></tr><tr><td>AC1-034</td><td>14.65</td><td>\$350,395</td></tr><tr><td>AC1-054</td><td>7.02</td><td>\$167,902</td></tr><tr><td>AC1-086</td><td>179.79</td><td>\$4,300,167</td></tr><tr><td>AC1-099</td><td>17.23</td><td>\$376,546</td></tr><tr><td>AC1-163</td><td>2.51</td><td>\$54,854</td></tr></table>	Queue	Impact (MW)	Cost	AB2-169	7.59	\$181,536	AC1-034	14.65	\$350,395	AC1-054	7.02	\$167,902	AC1-086	179.79	\$4,300,167	AC1-099	17.23	\$376,546	AC1-163	2.51	\$54,854	Pending	\$5,000,000	\$54,854
Queue	Impact (MW)	Cost																								
AB2-169	7.59	\$181,536																								
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AC1-099	17.23	\$376,546																								
AC1-163	2.51	\$54,854																								
Total New Network Upgrades					\$54,854																					

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which is calculated and reported for in the Impact Study)

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AC1-163 Allocation
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Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AC1-163 Allocation																					
# 2	3BTLEBRO-3ROCKYMT115T 115 kV line	Replace Battleboro substation terminal equipment. Note: Duke/Progress Energy portion of this line will need to be studied under Duke’s FERC tariff process. <table><tr><th>Queue</th><th>Impact (MW)</th><th>Cost</th></tr><tr><td>AB2-059</td><td>24.577</td><td></td></tr><tr><td>AB2-100</td><td>7.96</td><td></td></tr><tr><td>AB2-174</td><td>6.80</td><td></td></tr><tr><td>AC1-034</td><td>13.41</td><td></td></tr><tr><td>AC1-086</td><td>21.59</td><td></td></tr><tr><td>AC1-163</td><td>1.7</td><td></td></tr></table>	Queue	Impact (MW)	Cost	AB2-059	24.577		AB2-100	7.96		AB2-174	6.80		AC1-034	13.41		AC1-086	21.59		AC1-163	1.7		Pending	\$15,000	\$0
Queue	Impact (MW)	Cost																								
AB2-059	24.577																									
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AC1-086	21.59																									
AC1-163	1.7																									
Total New Network Upgrades					\$0																					

Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The IC can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this interconnection request by addressing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

#	Contingency		Affected Area	Facility Description	Bus		Circuit	Power Flow	Loading %		Rating		MW Contribution
	Type	Name			From	To			Initial	Final	Type	MVA	

#	Contingency		Affected Area	Facility Description	Bus		Circuit	Power Flow	Loading %		Rating		MW Contribution
	Type	Name			From	To			Initial	Final	Type	MVA	
3	N-1	LN 246_B	DVP - DVP	6CLUBHSE-6SAPONY 230 kV line	314563	314435	1	AC	99.61	100.29	ER	599	4.15
4	N-1	3CAROLN A-6CAROLN A	DVP - DVP	3COX DP-3WITAKRS 115 kV line	314577	314623	1	AC	96.88	98.14	ER	134	1.7
5	N-1	3CAROLN A-6CAROLN A	DVP - DVP	3WITAKRS-3BTLEBRO 115 kV line	314623	314554	1	AC	95.63	96.89	ER	134	1.7

Light Load Analysis in 2020

Not required

ITO Analysis

ITO assessed the impact of the proposed Queue Project #AC1-163 interconnection of 16.9 MW of energy (Capacity 12.6 MW) for compliance with reliability criteria on ITO's Transmission System. The system was assessed using the summer 2020 RTEP case provided to ITO by PJM. When performing a generation analysis, ITO's main analysis will be load flow study results under single contingency and multiple facility contingency (both normal and stressed system conditions). ITO Criteria considers a transmission facility overloaded if it exceeds 94% of its emergency rating under normal and stressed system conditions. A full listing of ITO's Planning Criteria and interconnection requirements can be found in the ITO's Facility Connection Requirements which are publicly available at: <http://www.dom.com>.

The results of these studies evaluate the system under a limited set of operating conditions and do not guarantee the full delivery of the capacity and associated energy of this proposed interconnection request under all operating conditions. NERC Planning and Operating Reliability Criteria allow for the re-dispatch of generating units to resolve projected and actual deficiencies in real time and planning studies. Specifically NERC Category C Contingency Conditions (Bus Fault, Tower Line, N-1-1, and Stuck Breaker scenarios) allow for re-dispatch of generating units to resolve potential reliability deficiencies. For ITO Planning Criteria the re-dispatch of generating units for these contingency conditions is allowed as long as the projected loading does not exceed 100% of a facility Load Dump Rating.

As part of its generation impact analysis ITO routinely evaluates the impact that a proposed new generation resource will have under maximum generation conditions, stress system conditions and import/export system conditions (greater than 20 MW). The results of these studies are discussed in more detail below.

Category B Analysis (Single Contingency):

- System Normal – No deficiencies identified
- Critical System Condition (No Surry 230 kV Unit) OR (Possum Point Unit #6 (230kV Unit) – No deficiencies identified.

Category C Analysis: (Multiple Facility Contingency)

- Bus Fault - No deficiencies identified
- Line Stuck Breaker - No deficiencies identified
- Tower Line – No deficiencies identified

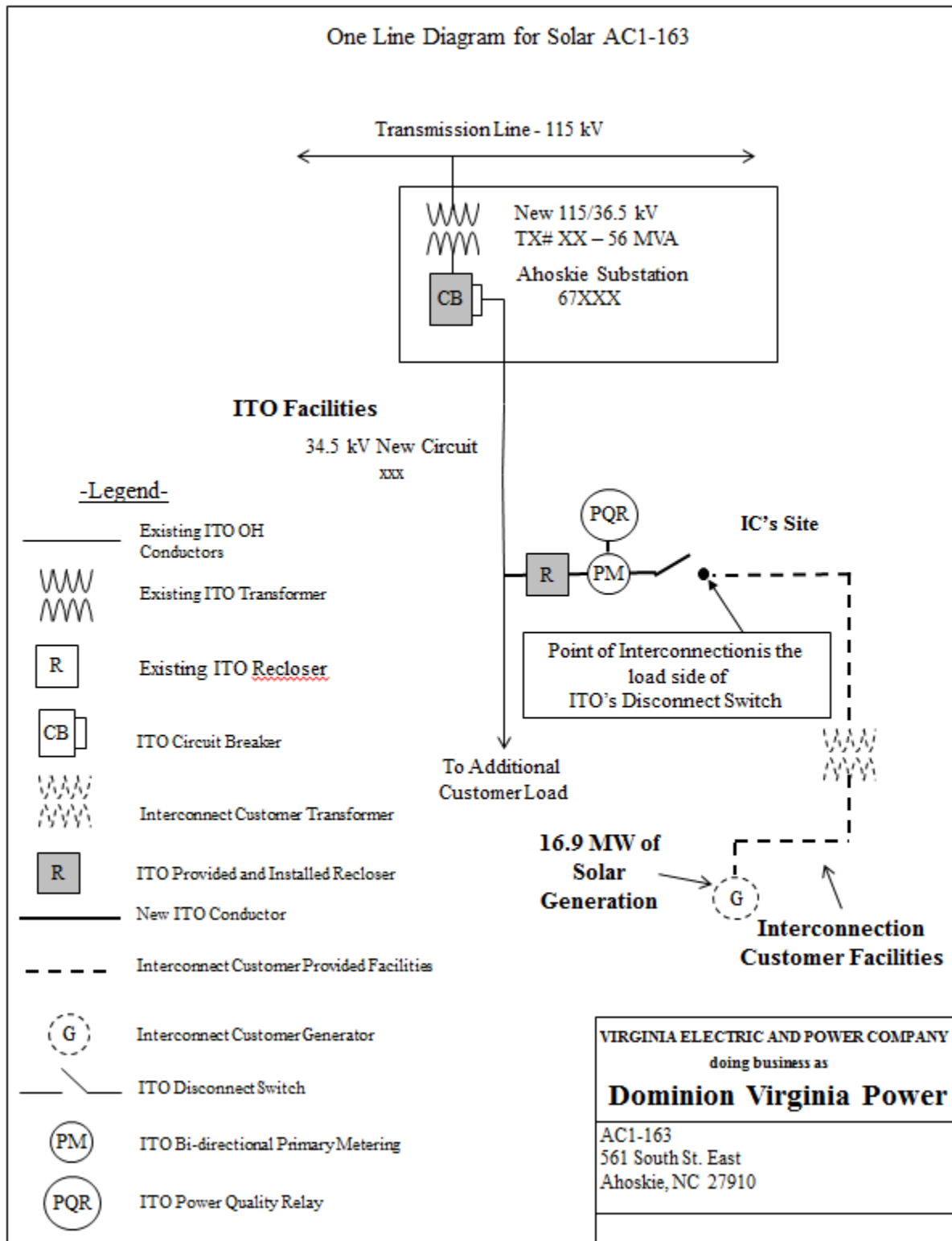
Affected System Analysis & Mitigation

Duke Energy:

Enter into an Affected System Facilities Study agreement with Duke / Progress Energy (DEP) to determine how to mitigate Battleboro - Rocky Mt.115kV overload. The upgrade will likely be a complete reconductor, probably replacing structures. The estimated cost \$15million with an ISD of 6/1/2020.

Attachment 1.

System Configuration



Appendices

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gage other generators impact.

It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

Appendix 1

(DVP - DVP) The AB2-100 TAP-6CLUBHSE 230 kV line (from bus 924510 to bus 314563 ckt 1) loads from 99.91% to 100.57% (AC power flow) of its emergency rating (375 MVA) for the single line contingency outage of 'LN 246_A'. This project contributes approximately 2.51 MW to the thermal violation.

CONTINGENCY 'LN 246_A'

OPEN BRANCH FROM BUS 921571 TO BUS 314575 CKT 1 /* AA1-138 TAP -
6NUCO TP 230.00

OPEN BRANCH FROM BUS 314569 TO BUS 314575 CKT 1 /* 6EARLEYS
230.00 - 6NUCO TP 230.00

OPEN BRANCH FROM BUS 314575 TO BUS 314590 CKT 1 /* 6NUCO TP
230.00 - 6NUCOR 230.00

OPEN BUS 314575 /* ISLAND

OPEN BUS 314590 /* ISLAND

END

Bus Number	Bus Name	Full Contribution
315292	1DOMTR78	1.36
315131	1EDGECEMA	2.01
315132	1EDGECEMB	2.01
315139	1GASTONA	7.82
315141	1GASTONB	7.82
315158	1KERR 1	0.28
315162	1KERR 5	0.16
315163	1KERR 6	0.16
315164	1KERR 7	0.16
315126	1ROARAP2	1.74
315128	1ROARAP4	1.67
315134	1ROAVALA	10.73
315135	1ROAVALB	2.86
315136	1ROSEMG1	5.25
315138	1ROSEMG2	2.46
315137	1ROSEMS1	3.26
900671	V4-068 C	0.12
917331	Z2-043 C	0.52
917341	Z2-044 C	0.28
917511	Z2-088 C OP1	0.98
917591	Z2-099 C	0.21
918411	AA1-050	0.83
LTF	AA1-058	0.5

921162	AA1-063AC	9.48
918511	AA1-065 C OP	2.11
921182	AA1-067 C	0.33
918561	AA1-072 C	0.08
921752	AA2-053 C	9.85
921762	AA2-057 C	7.82
921772	AA2-059 C	1.04
921862	AA2-068 C	2.48
920021	AA2-086 C	0.11
921982	AA2-088 C	7.21
922442	AA2-165 C	1.07
922472	AA2-169 C	1.52
922512	AA2-174 C	0.45
922532	AA2-178 C	5.99
922602	AB1-013 C	1.81
922922	AB1-081 C OP	9.73
923262	AB1-132 C OP	31.41
923941	AB2-035 C	0.37
924151	AB2-059 C	11.47
924381	AB2-087 C	0.68
924391	AB2-088 C	0.48
924401	AB2-089 C	1.07
924491	AB2-098 C	0.58
924501	AB2-099 C	0.7
924511	AB2-100 C	42.97
925121	AB2-169 C	1.11
925281	AB2-186 C	0.27
925291	AB2-188 C OP	1.48
926331	AC1-034 C OP	7.53
926531	AC1-054 C OP	4.11
926851	AC1-086 C	46.26
926981	AC1-099 C	9.28
927731	AC1-163 C	2.51
927991	AC1-189 C	8.93
928191	AC1-208 C O1	10.05

Appendix 2

(DVP - CPLE) The 3BTLEBRO-3ROCKYMT115T 115 kV line (from bus 314554 to bus 304223 ckt 1) loads from 156.84% to 157.85% (AC power flow) of its emergency rating (164 MVA) for the tower line contingency outage of 'LN 2058-2181'. This project contributes approximately 1.7 MW to the thermal violation.

CONTINGENCY 'LN 2058-2181'

OPEN BUS 304226 /* ISLAND: 6PA-RMOUNT#4115.00

OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1 /* 6PA-
RMOUNT#4230.00 - 6NASH 230.00

OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1 /* 6HATHAWAY
230.00 - 6NASH 230.00

OPEN BUS 314591 /* ISLAND: 6NASH 230.00

OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1 /*
6ROCKYMT230T230.00 - 6HATHAWAY 230.00

END

Bus Number	Bus Name	Full Contribution
315131	1EDGECSMA	2.59
315132	1EDGECSMB	2.59
315139	1GASTONA	2.51
315141	1GASTONB	2.51
315126	1ROARAP2	1.08
315128	1ROARAP4	1.04
315134	1ROAVALA	3.56
315135	1ROAVALB	0.95
315136	1ROSEMG1	2.02
315138	1ROSEMG2	0.95
315137	1ROSEMS1	1.25
900671	V4-068 C	0.07
900672	V4-068 E	0.19
917331	Z2-043 C	0.38
917332	Z2-043 E	0.86
917341	Z2-044 C	0.56
917342	Z2-044 E	1.26
917511	Z2-088 C OP1	0.73
917512	Z2-088 E OP1	6.22
917591	Z2-099 C	0.12
917592	Z2-099 E	0.28
918411	AA1-050	0.61
LTF	AA1-055	9.32

921162	AA1-063AC	5.73
921163	AA1-063AE	2.7
918512	AA1-065 E OP	2.03
921183	AA1-067 E	0.32
918561	AA1-072 C	0.06
918562	AA1-072 E	0.14
921752	AA2-053 C	5.82
921753	AA2-053 E	2.5
921762	AA2-057 C	13.08
921763	AA2-057 E	6.54
921862	AA2-068 C	3.37
921863	AA2-068 E	1.55
920021	AA2-086 C	0.06
920022	AA2-086 E	0.15
921982	AA2-088 C	4.15
921983	AA2-088 E	6.78
922442	AA2-165 C	1.78
922443	AA2-165 E	0.86
922472	AA2-169 C	0.92
922473	AA2-169 E	0.42
922512	AA2-174 C	0.27
922513	AA2-174 E	0.29
922922	AB1-081 C OP	20.25
922923	AB1-081 E OP	8.68
923262	AB1-132 C OP	10.08
923263	AB1-132 E OP	4.32
923572	AB1-173 C OP	1.29
923573	AB1-173 E OP	0.6
923582	AB1-173AC OP	1.29
923583	AB1-173AE OP	0.6
923911	AB2-031 C OP	1.28
923912	AB2-031 E OP	0.63
923941	AB2-035 C	0.38
923942	AB2-035 E	0.16
923991	AB2-040 C OP	4.2
923992	AB2-040 E OP	3.44
924151	AB2-059 C	23.86
924152	AB2-059 E	12.29
924381	AB2-087 C	0.34
924382	AB2-087 E	0.16
924391	AB2-088 C	0.48
924392	AB2-088 E	0.23
924491	AB2-098 C	0.25
924492	AB2-098 E	0.11
924501	AB2-099 C	0.35

924502	AB2-099 E	0.15
924511	AB2-100 C	5.52
924512	AB2-100 E	2.72
925122	AB2-169 E	2.31
925171	AB2-174 C OP	3.78
925172	AB2-174 E OP	3.42
926331	AC1-034 C OP	7.64
926332	AC1-034 E OP	5.76
926851	AC1-086 C	14.84
926852	AC1-086 E	6.75
926981	AC1-099 C	10.95
926982	AC1-099 E	6.5
LTF	AC1-133	8.97
927731	AC1-163 C	1.27
927732	AC1-163 E	0.44
927991	AC1-189 C	5.28
927992	AC1-189 E	2.63
928171	AC1-206 C OP	4.45
928172	AC1-206 E OP	2.12
928191	AC1-208 C O1	12.83
928192	AC1-208 E O1	5.7